



Saksham Raj
Computer Science & Engineering
Indian Institute of Technology Bombay

22B0964
B.Tech.
Gender: Male
DOB: 09/05/2004

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2026	c
Intermediate	CBSE	Subhash Public School, Jharkhand	2022	
Matriculation	CBSE	B.N.S. D.A.V. Public School, Jharkhand	2020	

SCHOLASTIC ACHIEVEMENTS

- Achieved **AIR 145** in the Joint Entrance Examination Main outperforming **1 million+** candidates ('22)
- Secured **All India Rank 323** in Joint Entrance Examination Advanced among **150,000+** candidates ('22)

INTERNSHIP EXPERIENCE

Software Engineering Intern

(May '25 - July '25)

Citicorp Services India Private Limited

- Developed a software platform to perform large computations in minutes, cutting manual effort from days
- Implemented an optimized Ordinary Least Squares (**OLS**) algorithm using **QR optimization**, also **DW and BP Test** from scratch, significantly enhancing the speed and efficiency of complex data processing
- Designed and integrated **Django-based APIs** to help streamline data workflows and enhance scalability
- Designed a **user-friendly UI** with added features delivering visually appealing and professional outputs

KEY PROJECTS

Automatic Index Creator in PostgreSQL Extension

(Mar '25 - Apr '25)

Prof. S. Sudarshan | CSE Dept. | DBIS Course Project

IIT Bombay

- Built an extension that hooks into **planner** to detect **sequential scans** and identify indexing opportunities
- Applied **SPI** (Server Programming Interface) for executing DDL operations from within the extension codebase and advanced internal PostgreSQL APIs like the **GUC**(Grand Unified Configuration) variables
- Designed a **cost-benefit analysis** using planner estimates to determine when index creation is justified
- Used **background workers** to create indexes asynchronously, avoiding disruption to query performance
- Maintained a table `aidx_queries` to monitor indexing candidates and query frequency for cost-benefit analysis

Compiler for SCLP

(Jan '25 - Mar '25)

Prof. Uday Khedker | CSE Dept. | Implementation of Programming Languages Course Project

IIT Bombay

- Built a **compiler** for **C++**-like language designed with a clean, class-based and object-oriented structure
- Implemented an end-to-end compilation pipeline covering the major stages of **AST** generation, **TAC** creation, **RTL** emission, and the final optimized assembly-like code output, enabling a smooth transformation
- Developed memory management and register allocation strategies to improve performance of generated code

Audio-Based MAC Protocol

(Sep '24 - Oct '24)

Prof. Vinay Ribeiro | CSE Dept. | Computer Networks Course Project

IIT Bombay

- Developed a custom **MAC**-layer protocol with **RTS** control signaling and synchronization to manage multi-node transmissions, prevent collisions, and guarantee exclusive channel use during message delivery
- Designed a scheduling system for unicast and broadcast delivery with automatic retransmission on failures
- Coordinated transmission timing across nodes to maintain sequence and prevent message transfer overlaps
- Added payload validation, binary address encoding, and end markers to guarantee reliable communication

Handcrafted JPEG: Image Compression

(Sep '24 - Nov '24)

Prof. Ajit Rajwade | CSE Dept. | Fundamentals of Digital Image Processing Course Project

IIT Bombay

- Built custom JPEG encoder & decoder from scratch using Python, showcasing image compression expertise
- Used discrete cosine transform(**DCT**), quantization, **Huffman coding**, zig-zag scanning & **RLE** on zeros
- Introduced an **optimization** by omitting Huffman code lengths, exploiting the **prefix-free** property, thereby achieving the compression ratios close to standard JPEG for select images at the same quality factor
- Reconstructed encoded images using **IDCT**, de-quantization & Huffman decoding to obtain accurate image

Bus Reservation System

(May '25 - Jun '25)

Self Project

IIT Bombay

- Built a C++ bus reservation system with PostgreSQL integration using **libpqxx** for robust data handling
- Designed **atomic** booking and cancellation workflows ensuring real-time seat updates and data consistency
- Created a modular, **menu-driven** interface for intuitive navigation and scalable feature expansion in future

Full-Stack E-Commerce Web Application

(Jan '25 - Feb '25)

Prof. S. Sudarshan | CSE Dept. | DBIS Course Project

IIT Bombay

- Engineered a modular full-stack e-commerce platform using **Node.js + Express** for REST API backend, **React.js** for frontend, and **PostgreSQL** for persistent data storage, ensuring clean separation of concerns
- Built secure authentication with **bcrypt** hashing and cross-origin credential handling for persistent logins
- Implemented end-to-end e-commerce functionality: product listing with search and quantity controls, shopping cart management, order placement with stock validation, and address auto-fill via **Postal API**

Algorithmic Trader

(Oct '23 - Nov '23)

Prof. Ashutosh Gupta | CSE Dept. | DSA Course Project

IIT Bombay

- Developed an algorithmic trading system in **C++**, integrating data structures like hash tables & linked lists
- Applied **OOP** principles to create modular and scalable code thus enhancing maintainability and readability
- Effectively implemented trading strategies like buying low/selling high & detecting **arbitrage** opportunities

Operating System Snippets

(Jan '24 - Apr '24)

Prof. Mythili Vutukuru | CSE Dept. | OS Course Project

IIT Bombay

- Developed a custom shell with functionalities like **background/foreground** execution, **parallel/serial** command handling, **signal** management, **process** group control and **background process** management
- Implemented a simple file system emulator, including functionalities for file **creation**, deletion, **opening**, closing, **reading**, **writing** and offset management utilizing **block-based storage** and **inode** management

Hybrid Minesweeper and Cricket Game

(Mar '23 - June '23)

Prof. Kameswari Chebrolu | CSE Dept. | Software Systems Lab Course Project

IIT Bombay

- Successfully coded and deployed a web-based game which blends 2 unique games, Minesweeper and Cricket
- Proficiently employed HTML, CSS, and JavaScript to establish a solid and reliable foundation of the game
- Developed Javascript algorithms to seamlessly manage a dynamic 6×6 game grid housing 11 concealed fielders, orchestrating their **position changes on each turn**, to enhance the overall gameplay complexity
- Implemented a **versatile scoring system**, introducing multiple score types based on placement of fielders

Algorithmic Machine Learning

(Jan '24 - Apr '24)

Prof. Swaprava Nath | CSE Dept. | AI/ML Course Project

IIT Bombay

- Developed machine learning algorithms from scratch in Python (**Linear Regression**, Logistic Regression, **Naive Bayes**, Decision Trees and Perceptrons), demonstrating strong understanding of ML fundamentals
- Implemented NNs without the libraries (Feedforward NNs, **CNNs**) showcasing proficiency in Deep Learning
- Applied mathematical concepts via implementation of Softmax, **gradient descent** and **backpropagation**

Algorithmic Trading (Arbitrage)

(Jun '23 - Jul '23)

Summer of Science | Maths and Physics Club

IIT Bombay

- Conducted an in-depth study on algorithmic trading principles, with a primary focus on **arbitrage** strategies, particularly quantitative methods such as Statistical Arbitrage, Autoregression, and Dynamic Linear Models
- Analyzed market dynamics like interstock volatility and assessed real-time trading implementation challenges

TECHNICAL SKILLS

Programming	Bash C/C++ Python MIPS Assembly SQL JavaScript
Development	Django HTML CSS React Make NodeJS ExpressJS
Softwares and Packages	GitHub L ^A T _E X NumPy Pandas Matplotlib python-pptx

EXTRACURRICULAR ACHIEVEMENTS

- Led a group of 7 on the idea of Implementing **IoT** in Traffic Management system, did customer discovery for **hypothesis validation** and also worked on the major aspects aligning the **Business Model Canvas**
- Crafted the business model, identified its differentiation, understood the technology, and analyzed financials for a Blockchain startup idea. Led a team of 3 to present it at the EnB Buzz Business Pitching competition
- Volunteered at Green Campus—NSS, preparing reports, planting trees, and learning about the local flora